Solutions to Topology Problems 6–10

Throughout, a topology on a set X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ such that (1) $\varnothing, X \in \mathcal{T}$, (2) arbitrary unions of members of $\mathcal{T}$ belong to $\mathcal{T}$, and (3) finite intersections of members of $\mathcal{T}$ belong to $\mathcal{T}$.

6. The collections $\mathcal{T}_1$ and $\mathcal{T}_2$ on $\mathbb{N}$

Let

$$A_n = \{1, 2, \ldots, n\}, \qquad B_n = \mathbb{N} \setminus A_n \quad (n \geq 1).$$

Then the two collections in the question can be written as

$$\mathcal{T}_1 = \{\varnothing, \mathbb{N}\} \cup \{A_n : n \geq 1\}, \qquad \mathcal{T}_2 = \{\varnothing, \mathbb{N}\} \cup \{B_n : n \geq 1\}.$$

We verify the topology axioms separately.

(a) The collection $\mathcal{T}_1$ is a topology. Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_1$. For finite intersections, observe that

$$A_m \cap A_n = A_{\min\{m,n\}}.$$

Intersections with $\varnothing$ or $\mathbb{N}$ also belong to $\mathcal{T}_1$. Thus $\mathcal{T}_1$ is closed under finite intersections.

Now let $\{U_i\}_{i \in I}$ be any family of members of $\mathcal{T}_1$. If every U_i is empty, then $\bigcup_{i \in I} U_i = \varnothing$. If at least one $U_i = \mathbb{N}$, the union is $\mathbb{N}$. In the remaining case, the nonempty sets in the family are of the form A_n. If their indices are unbounded in $\mathbb{N}$, then every positive integer belongs to some A_n, so their union is $\mathbb{N}$. If their indices are bounded, they have a largest value m (a nonempty bounded subset of $\mathbb{N}$ has a maximum), and the union is A_m. In every case the union belongs to $\mathcal{T}_1$. Hence $\mathcal{T}_1$ is a topology on $\mathbb{N}$.

(b) The collection $\mathcal{T}_2$ is a topology. Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_2$. For finite intersections, De Morgan's law gives

$$B_m \cap B_n = (\mathbb{N} \setminus A_m) \cap (\mathbb{N} \setminus A_n) = \mathbb{N} \setminus (A_m \cup A_n) = B_{\max\{m,n\}}.$$

Intersections involving $\varnothing$ or $\mathbb{N}$ also remain in $\mathcal{T}_2$.

Consider an arbitrary family $\{U_i\}_{i \in I} \subseteq \mathcal{T}_2$. If one member is $\mathbb{N}$, its union is $\mathbb{N}$. If every member is empty, its union is empty. Otherwise, discard the empty members and write the remaining members as B_{n_i}. The set of indices $\{n_i : i \in I,\ U_i \neq \varnothing\}$ is a nonempty subset of $\mathbb{N}$, so it has a least element, say m. Since the sets B_n increase as n decreases, we have

$$\bigcup_{i \in I} B_{n_i} = B_m.$$

Indeed, B_m is one of the sets in the union and contains every B_{n_i} because $m \leq n_i$. Thus the union belongs to $\mathcal{T}_2$. Therefore $\mathcal{T}_2$ is also a topology on $\mathbb{N}$.

Conclusion: Both $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

7. The topology generated by a basis

Let $\mathcal{B}$ be a basis for a topology on X, and let $\mathcal{T}_{\mathcal{B}}$ denote the topology generated by $\mathcal{B}$. By the definition of the topology generated by a basis,

$$\mathcal{T}_{\mathcal{B}} = \left\{ \bigcup_{\alpha \in A} B_\alpha : A \text{ is an index set and } B_\alpha \in \mathcal{B} \right\}.$$

Let $\mathfrak{F}$ be the collection of all topologies on X that contain $\mathcal{B}$. We prove

$$\boxed{\mathcal{T}_{\mathcal{B}} = \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.}$$

First, let $U \in \mathcal{T}_{\mathcal{B}}$. Then U is a union of elements of $\mathcal{B}$. If $\mathcal{T} \in \mathfrak{F}$, then $\mathcal{B} \subseteq \mathcal{T}$. Since every topology is closed under arbitrary unions, it follows that $U \in \mathcal{T}$. This holds for every $\mathcal{T} \in \mathfrak{F}$, and hence

$$\mathcal{T}_{\mathcal{B}} \subseteq \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.$$

For the reverse inclusion, note that $\mathcal{T}_{\mathcal{B}}$ is itself a topology on X containing $\mathcal{B}$. Consequently, $\mathcal{T}_{\mathcal{B}} \in \mathfrak{F}$. The intersection of all members of $\mathfrak{F}$ is therefore contained in this particular member:

$$\bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T} \subseteq \mathcal{T}_{\mathcal{B}}.$$

Combining the two inclusions proves the required equality.

Interpretation: The topology generated by a basis is the smallest topology on X that contains every basis element.

8. Nonempty open sets in the finite-complement topology

Let X be infinite and let

$$\mathcal{T}_f = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$$

be the finite-complement topology. Suppose $U, V \in \mathcal{T}_f$ are nonempty. By the definition of $\mathcal{T}_f$, both complements $X \setminus U$ and $X \setminus V$ are finite. Using De Morgan's law,

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V).$$

The right-hand side is a union of two finite sets, and is therefore finite. If $U \cap V = \varnothing$, then the left-hand side equals X, which would imply that X is finite. This contradicts the assumption that X is infinite. Hence

$$\boxed{U \cap V \neq \varnothing.}$$

9. The particular-point topology

Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a = \{V \subseteq X : V = \varnothing \text{ or } a \in V\}.$$

We check the topology axioms.

(i) Empty set and whole space. By definition, $\varnothing \in \mathcal{T}_a$. Also $a \in X$, so $X \in \mathcal{T}_a$.

(ii) Arbitrary unions. Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a$. If every V_i is empty, then $\bigcup_{i \in I} V_i = \varnothing$. Otherwise, at least one member, say V_j, is nonempty. Since $V_j \in \mathcal{T}_a$, it contains a. Therefore $a \in \bigcup_{i \in I} V_i$, so the union belongs to $\mathcal{T}_a$.

(iii) Finite intersections. It is enough to check intersections of two members, since repeated application gives the result for every finite intersection. Let $U, V \in \mathcal{T}_a$. If either set is empty, then $U \cap V = \varnothing \in \mathcal{T}_a$. If neither is empty, both contain a, and hence $a \in U \cap V$. Thus $U \cap V \in \mathcal{T}_a$.

All three axioms hold. Therefore $\boxed{\mathcal{T}_a \text{ is a topology on } X.}$

10. The excluded-point topology

Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a^e = \{V \subseteq X : V = X \text{ or } a \notin V\}.$$

Thus the open sets are X itself and all subsets of X that do not contain a. We verify the topology axioms.

(i) Empty set and whole space. Since $a \notin \varnothing$, we have $\varnothing \in \mathcal{T}_a^e$. Also $X \in \mathcal{T}_a^e$ by definition.

(ii) Arbitrary unions. Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a^e$. If some $V_j = X$, then $\bigcup_{i \in I} V_i = X$. If no member equals X, then every V_i omits a. Consequently, their union also omits a:

$$a \notin V_i \text{ for all } i \in I \quad \Longrightarrow \quad a \notin \bigcup_{i \in I} V_i.$$

In either case the union belongs to $\mathcal{T}_a^e$.

(iii) Finite intersections. Again it is enough to consider two sets. Let $U, V \in \mathcal{T}_a^e$. If both equal X, then $U \cap V = X$. Otherwise, at least one of them, say U, is not X. By the definition of the collection, $a \notin U$. Since $U \cap V \subseteq U$, the intersection also omits a. Hence $U \cap V \in \mathcal{T}_a^e$.

All three topology axioms are satisfied. Therefore

$$\boxed{\mathcal{T}_a^e \text{ is a topology on } X.}$$